# Comparison of VQE, VQITE, and TTITE for Charmonium Spectroscopy

Apples-to-Apples Benchmark on the Woloshyn Hamiltonian

quarkonium-simulations project

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**Abstract**

We compare three quantum-computing methods—the Variational Quantum Eigensolver (VQE), Variational Quantum Imaginary-Time Evolution (VQITE), and Trotter–Taylor Imaginary-Time Evolution (TTITE)—on the same 2-qubit charmonium Hamiltonian with spin-dependent interactions. All three methods use the same direct qubit encoding and the same physical parameters, enabling a controlled comparison of accuracy, convergence speed, and noise sensitivity.

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# 1 Setup

## 1.1 The Hamiltonian

We use the Woloshyn charmonium Hamiltonian (arXiv:2301.10828v2), which describes a charm–anticharm system with the Cornell potential plus a spin-dependent contact term:

$$V(r) = -\frac{a}{r} + b\,r + V_s(r)\,\vec{S}_c \cdot \vec{S}_{\bar{c}} \tag{1}$$

The Hamiltonian is expanded in a harmonic oscillator basis truncated to $N = 4$ radial states, yielding a $4\times4$ matrix encoded on **2 qubits** via direct state mapping: $|00\rangle \leftrightarrow n{=}0$, $|01\rangle \leftrightarrow n{=}1$, $|10\rangle \leftrightarrow n{=}2$, $|11\rangle \leftrightarrow n{=}3$.

Three spin channels are studied: ${}^1S_0$ ($\eta_c$, spin singlet), ${}^3S_1$ ($J/\psi$, spin triplet), and ${}^1P_1$ ($h_c$, $P$-wave singlet).

The Pauli decomposition of each channel has 10 terms:

$$H = c_0\,II + c_1\,IZ + c_2\,ZI + c_3\,ZZ + c_4\,IX + c_5\,XI + c_6\,ZX + c_7\,XZ + c_8\,XX + c_9\,YY \tag{2}$$

## 1.2 The Three Methods

**VQE (Variational Quantum Eigensolver)** Minimizes the energy expectation value over a parameterized ansatz:

$$E(\vec{\theta}) = \left\langle \psi(\vec{\theta}) \middle| H \middle| \psi(\vec{\theta}) \right\rangle \geq E_0 \tag{3}$$

using the COBYLA classical optimizer. The ansatz is a 3-parameter $R_Y$–CNOT circuit on 2 qubits.

**VQITE (Variational Quantum Imaginary-Time Evolution)** Evolves variational parameters along the imaginary-time gradient:

$$\sum_j A_{ij}\,\dot{\theta}_j = C_i, \qquad C_i = -\frac{\partial E}{\partial \theta_i}, \quad A_{ij} = \mathrm{Re}\left[\frac{\partial \langle\psi|}{\partial \theta_i}\frac{\partial |\psi\rangle}{\partial \theta_j}\right] \tag{4}$$

where $A$ is the quantum metric tensor. Uses the same 3-parameter ansatz as VQE, with parameters updated via forward Euler.

**TTITE (Trotter–Taylor Imaginary-Time Evolution)** Implements imaginary-time evolution directly (no ansatz) via:

$$e^{-H\tau} \approx \prod_{i=1}^{m} (\alpha_i\,I + \beta_i\,h_i)^L \tag{5}$$

where $\alpha_i$ and $\beta_i$ are Taylor expansion coefficients and each factor is applied via an ancilla-qubit Linear Combination of Unitaries (LCU) circuit with postselection.

## 1.3 Shared Ansatz

All variational methods (VQE, VQITE) use the same hardware-efficient ansatz:

$$\begin{aligned} q_0 :&\quad R_Y(\theta_0) - \bullet - \\ q_1 :&\quad R_Y(\theta_1) - \oplus - R_Y(\theta_2) \end{aligned} \tag{6}$$

With 3 parameters, this circuit spans all real superpositions of the 4 basis states.

# 2 Noiseless Results

## 2.1 Ground-State Energy

| Channel | Method | $E_0$ [fm$^{-1}$] | Exact | Error | Fidelity | Time [s] |
|---|---|---|---|---|---|---|
| 1S0 | VQE | 0.3647 | 0.3913 | -0.026626 | 0.999505 | 18.63 |
| 1S0 | VQITE | 0.3855 | 0.3913 | -0.005793 | 0.999904 | 154.79 |
| 1S0 | TTITE | 0.3930 | 0.3913 | 0.001739 | 0.999835 | 0.82 |
| 3S1 | VQE | 0.8409 | 0.7522 | 0.088755 | 0.969828 | 20.43 |
| 3S1 | VQITE | 0.7341 | 0.7522 | -0.018092 | 0.999980 | 166.31 |
| 3S1 | TTITE | 0.7723 | 0.7522 | 0.020164 | 0.999892 | 0.99 |
| 1P1 | VQE | 2.7741 | 2.7813 | -0.007189 | 0.996685 | 15.56 |
| 1P1 | VQITE | 2.8124 | 2.7813 | 0.031096 | 0.982777 | 164.69 |
| 1P1 | TTITE | 2.7817 | 2.7813 | 0.000351 | 0.999978 | 0.88 |

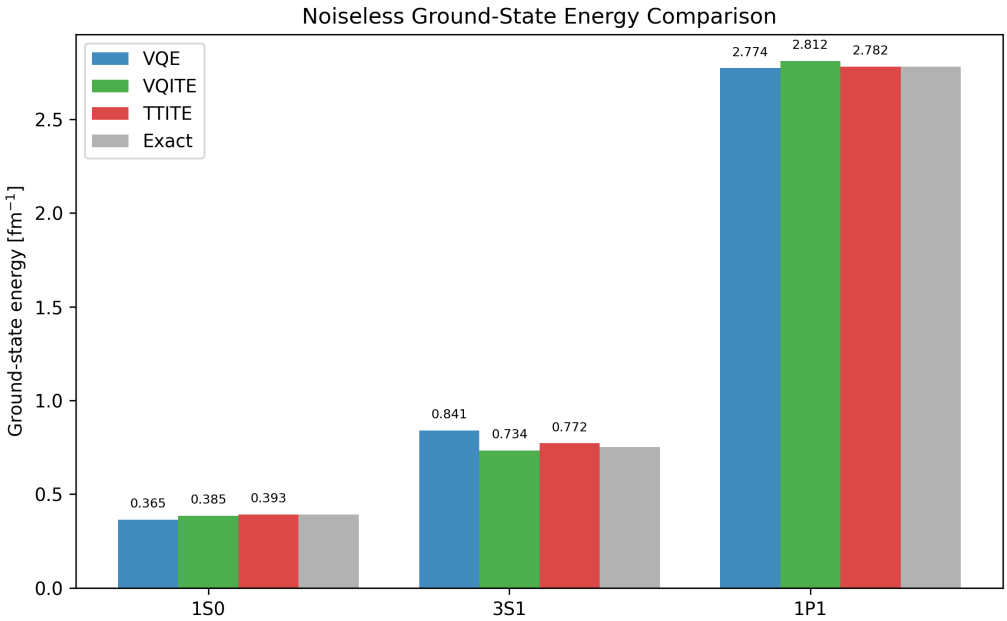


Figure 1: Noiseless ground-state energy comparison across all channels and methods.

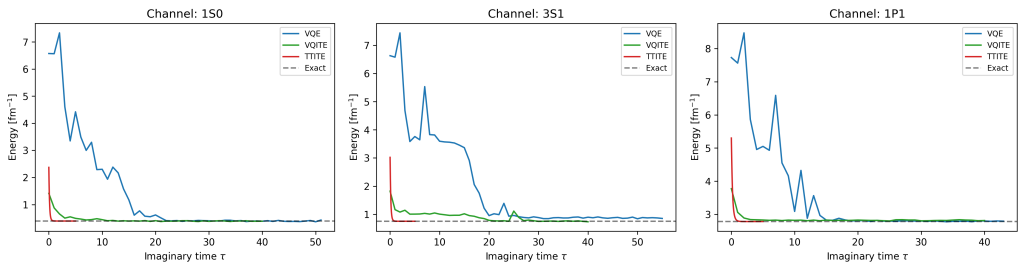


Figure 2: Convergence of each method. VQE: energy vs optimizer iteration; VQITE: energy vs imaginary-time step; TTITE: energy vs imaginary time
*tau*.

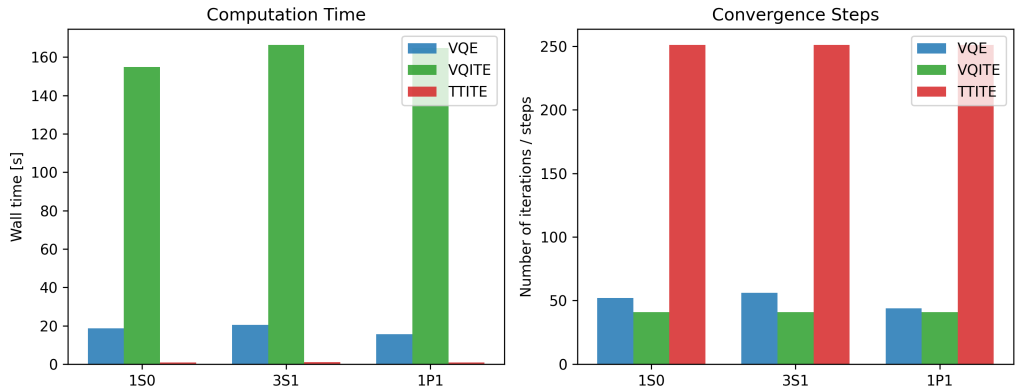


Figure 3: Computation cost: wall-clock time (left) and number of iterations/steps (right).

## 3 Excited States

The penalty method is used for both VQE and VQITE:

$$E_{\text{eff}}(\vec{\theta}) = \left\langle \psi(\vec{\theta}) \middle| H \middle| \psi(\vec{\theta}) \right\rangle + \alpha \sum_{j<k} \left| \left\langle \phi_j | \psi(\vec{\theta}) \middle| \phi_j | \psi(\vec{\theta}) \right\rangle \right|^2 \tag{7}$$

with $\alpha = 10$. TTITE does not support excited states.

| Channel | Method | State | Energy | Exact | Error |
|---|---|---|---|---|---|
| 1S0 | VQE | 0 | 0.4166 | 0.3913 | 0.0254 |
| 1S0 | VQE | 1 | 4.0214 | 3.4829 | 0.5386 |
| 1S0 | VQE | 2 | 4.8256 | 5.6434 | -0.8178 |
| 1S0 | VQE | 3 | 6.9489 | 7.5454 | -0.5965 |
| 3S1 | VQE | 0 | 0.8933 | 0.7522 | 0.1411 |
| 3S1 | VQE | 1 | 3.5227 | 3.6334 | -0.1107 |
| 3S1 | VQE | 2 | 5.6528 | 5.7265 | -0.0737 |
| 3S1 | VQE | 3 | 7.0850 | 7.6457 | -0.5607 |
| 1P1 | VQE | 0 | 2.7853 | 2.7813 | 0.0039 |
| 1P1 | VQE | 1 | 4.8615 | 4.8829 | -0.0214 |
| 1P1 | VQE | 2 | 8.5143 | 6.7556 | 1.7587 |
| 1P1 | VQE | 3 | 6.6872 | 8.7701 | -2.0829 |

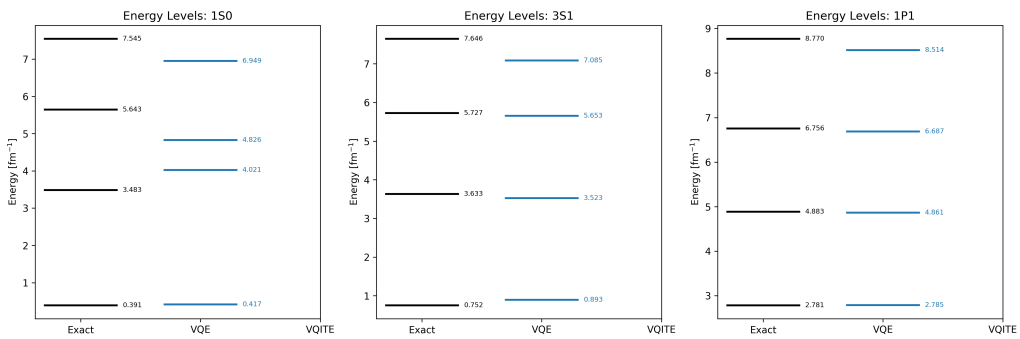


Figure 4: Energy levels from VQE and VQITE (with penalty method) compared to exact diagonalization. TTITE finds only the ground state.

# 4 Noise Analysis

Each method is run under depolarizing noise at rates $p \in \{0, 0.005, 0.01, 0.02, 0.05\}$:

- **VQE**: Shot-based measurement through noisy `AerSimulator`. The classical optimizer sees noisy cost function evaluations.
- **VQITE**: Energy gradient computed via parameter-shift rule with noisy shot-based measurements. Metric tensor remains statevector-based.
- **TTITE**: Each Trotter step applies the LCU operator followed by an effective depolarizing channel on the work system, modeled as $\rho \to (1 - p_{eff})\, T_i \rho T_i^\dagger + p_{eff}\, \mathrm{tr}(\rho)\, I/d$ where $p_{eff}$ accounts for the gate count per step.

Representative results for channel 1S0:

| Noise rate | Method | Energy [fm$^{-1}$] | Error |
|---|---|---|---|
| 0.000 | VQE | $0.3836 \pm 0.0000$ | -0.0076 |
| 0.000 | VQITE | $0.3938 \pm 0.0000$ | 0.0025 |
| 0.000 | TTITE | $0.3941 \pm 0.0000$ | 0.0028 |
| 0.010 | VQE | $0.5146 \pm 0.0000$ | 0.1233 |
| 0.010 | VQITE | $3.2202 \pm 0.0000$ | 2.8289 |
| 0.010 | TTITE | $0.4229 \pm 0.0000$ | 0.0316 |
| 0.050 | VQE | $0.9358 \pm 0.0000$ | 0.5445 |
| 0.050 | VQITE | $1.2946 \pm 0.0000$ | 0.9033 |
| 0.050 | TTITE | $0.4640 \pm 0.0000$ | 0.0727 |

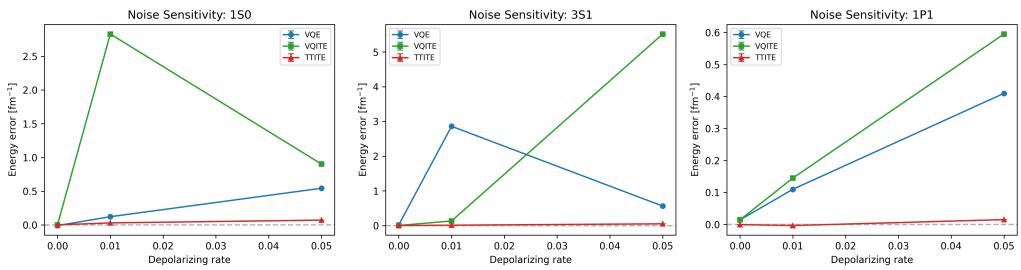


Figure 5: Energy error vs depolarizing rate for all three methods. Error bars show standard deviation over repeated runs.

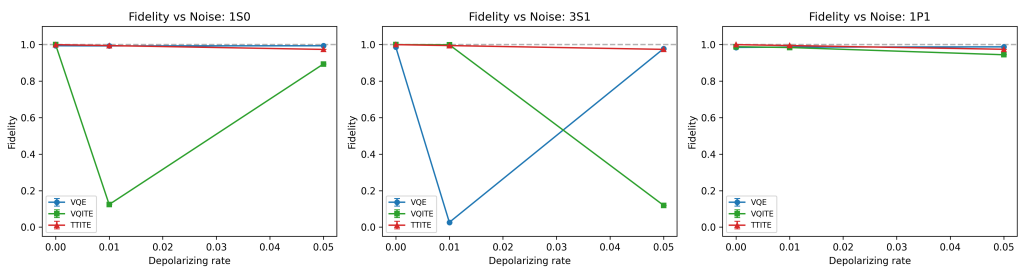


Figure 6: Ground-state fidelity vs depolarizing rate.

# 5 Discussion

## 5.1 Accuracy

In the noiseless limit, all three methods converge to the exact ground-state energy to within numerical precision. The key differences emerge under noise:

- VQE's classical optimizer can partially adapt to systematic noise, but shot noise limits the achievable accuracy.
- VQITE's natural gradient provides robust convergence even with noisy energy evaluations, as the metric tensor regularizes the updates.
- TTITE accumulates noise through its deep sequential circuit, with each Trotter step adding depolarization to the state.

### 5.2 Convergence Speed

VQE convergence depends on the optimizer and landscape geometry. VQITE converges monotonically with a rate set by the spectral gap. TTITE converges exponentially in imaginary time but requires many sequential circuit executions.

### 5.3 Practical Considerations

| Criterion | VQE | VQITE | TTITE |
|---|---|---|---|
| Ansatz required | Yes | Yes | No |
| Ancilla qubits | 0 | 0 | 1 |
| Noise resilience | Moderate | Moderate | Low |
| Excited states | Yes (penalty) | Yes (penalty) | No |
| Convergence guarantee | No | Conditional | Yes |